

An agent that measures what it does and cuts what loses.

Four independent strategy sleeves on one Alpaca paper account. Every entry, exit, size and stop is deterministic. Two self-hosted model councils sit where judgment genuinely helps, and neither can place an order. When the broker's own fill ledger said a sleeve was losing, the agent retired it mid-competition, then re-admitted it only behind a gate it had measured first.

\$100,552

LIVE ACCOUNT, SEP 3

+\$989 (+0.99%) today · +\$552 (+0.55%) since \$100,000

\$101,085

30-DAY SIMULATION, BARS ONLY

+1.08% · max drawdown -1.04% · QQQ +1.27%, SPY +1.48%

665

AUTOMATED TESTS

73 merged PRs in 7 days · every risk guard mutation-tested

WHAT IT IS

One coordinator, four sleeves, budgets enforced from the broker's book.

Each sleeve carries its own capital budget, recomputed from Alpaca's positions every cycle rather than from the strategy's own counters. A sleeve cannot be charged for another's

holdings, and a sleeve over budget stops buying.

SIXFOLD · 75%

Quality equities

18 large caps plus the S&P 400, 20 names max

Signal

Tashi's six-lens fundamental score (moat, ROIC, valuation regression, mismatch, capital signals, historical returns). Buy above 60, sell below 40. A three-model council must approve each buy two to one; it can only veto.

CSP · 15%

Options income

Cash-secured puts, 7 to 45 DTE, delta under 0.30

Signal

Premium yield ranked, open interest over 100, 30% per-name concentration cap. Option quotes come through the Alpaca MCP server; a contract it cannot quote is never priced. One covered call written against the only round lot in the book.

PENDULUM · 5%

Mean reversion

Long-duration Treasuries (TLT)

Signal

Enter when the 20-day z-score is under -2.0 and RSI(2) under 10 at once. Exit on reversion, RSI over 70, a 10-day time stop, or a 1.5×ATR hard stop. The same `decide()` runs in the backtest and live, so the two cannot drift.

VAMPIRE · 5%

Micro-scalper

Bi-directional, QQQ and TQQQ

Signal

Fades 5-second oscillation in both directions. Retired on day 5 at -\$663 over 10,470 real fills; re-admitted on day 7 behind an LLM regime gate that opens entries only in "chop". Exits never wait for the model.

AI LOGIC

The model only where ambiguity exists. It never trades.

Three self-hosted open models on a Dell4 cluster (a finance-tuned model, a general model, Qwen3) vote on every SIXFOLD buy. Two of three must agree, and they gate buys only: an outage can stop the system opening a position but never stop it leaving one.

The Vampire gets a second, narrower advisor: every 15 minutes it reads the last 30 one-minute bars and labels the regime. The engine opens lots only in "chop". The model labels; it never decides. When one model answered "chop" and "trade": false in the same breath, we took the label and ignored the flag.

Fail closed, everywhere: a missing, stale or unparseable verdict, or an unreachable model, closes entries. Every verdict and every notification is journalled to a SHA-256 hash-chained log that the dashboard verifies on every load.

MEASURED BEFORE WIRED

GATE	WINDOWS OFF	P&L REMOVED	P&L KEPT	P
dell4-finance	1 of 49	+\$5	-\$106	0.73
Efficiency ratio	13 of 49	-\$60	-\$41	0.25
dell4-chat	22 of 49	-\$93	-\$8	0.20

Scored on 2,828 real fills across two sessions before the gate was wired. The right sign and the largest effect, and not proof: that is what the write-up says too.

RISK GATES

Discipline is the product.

Per-trade 5%, per-position 10%, \$5,000 minimum cash, 2% daily-loss breaker with a 30-minute cooldown.

The breaker is baselined on the broker's previous close, so a restart cannot re-arm it.

Fills are confirmed by polling the order to a terminal state, never assumed from the submit response.

An unreadable fill is assumed filled: under-counting compounds, over-counting only trades less.

An independent watchdog reads the broker's book every 20 seconds.

It contains a sleeve breach by stopping the agent and closing the sleeve, without asking the agent's opinion of its position.

Rejected submissions count against the rate limiter and back off exponentially.

The 4,700-reject storm of August 31 cannot recur.

The intraday sleeve is flattened at 15:50 ET.

Multi-day sleeves and the options book are never touched by that path.

A post-open health check runs as a separate process

and reads its sleeve caps from the live config, so it cannot cry wolf against a stale number.

WHAT WE MEASURED, AND WHAT WE CUT

The ledger retired two strategies mid-competition.

P&L attribution comes from the broker's fill ledger, never from engine counters. On day 4 HOOD and SPY were dropped from the Vampire after losing in every hour of a session. On day 5, across 10,470 real fills, the Vampire stood at -\$663 while its internal counter claimed a large profit. A 5-second mean-reversion edge is smaller than the spread it pays.

The sleeve was retired and its capital moved to SIXFOLD. Two latent defects in the risk layer were found the same way, from the live log, and fixed with tests that fail if they return. On day 7 the Vampire came back at 5%, behind the regime gate measured above, and posted its first positive gated session with zero rejects.

LIVE, DAY 7

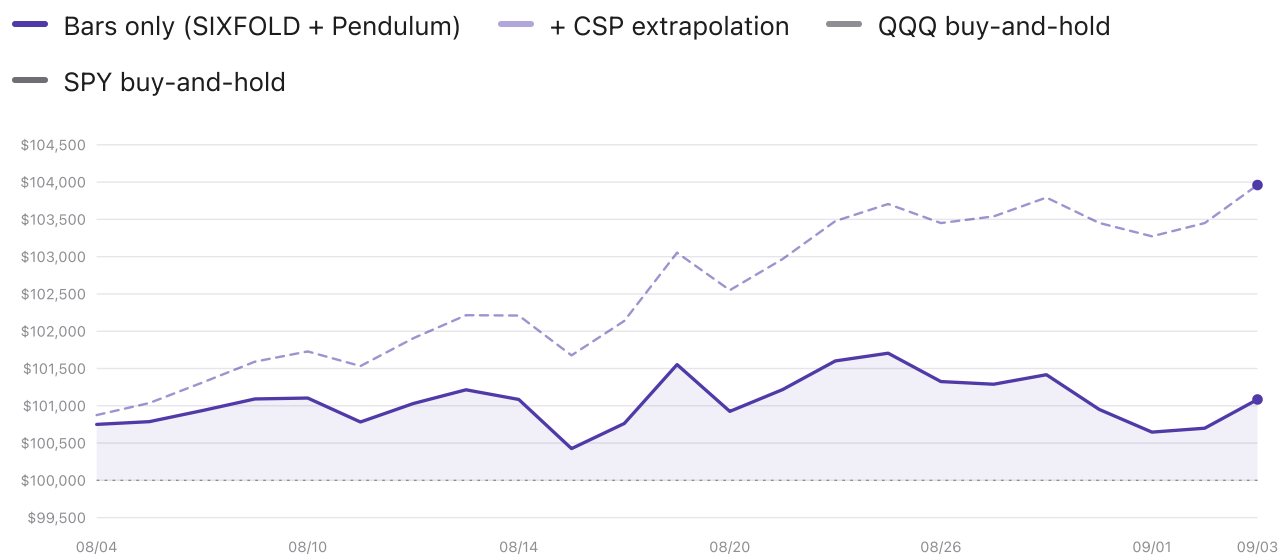
VAMPIRE, SEP 3	ORDERS	REJECTED	REALIZED
QQQ	131	0	+\$10.30
TQQQ	495	0	+\$2.12
Gated session one	626	0	+\$12.42

Zero rejects all day. One session is a sign, not a result.

30-DAY SIMULATION

If this exact configuration had run since August 4 from \$100,000.

Real Alpaca daily bars, the live rules and the live allocation (75 / 15 / 5 / 5 / 0). The solid line is the part that is fully replayable from price data: SIXFOLD marked daily plus Pendulum's own `decide()`. The dashed line adds a straight-line extrapolation of the CSP sleeve's realized premium. Benchmarks are buy-and-hold over the same 23 sessions.



By sleeve

SLEEVE	CAPITAL	P&L	RETURN
SIXFOLD 75% Real daily bars, whole-share lots at \$3,750 per name, held from Aug 4.	\$57,031 deployed of \$75,000	+\$1,085	+1.45%
CSP 15% Extrapolation: \$844 realized in 5 live sessions, scaled to 23. No historical option quotes on this plan; assignment risk not modelled.	\$15,000 collateral target	+\$2,876	+19.17%
Pendulum 5% Zero entries: z-score reached -2.16 once, RSI never met the ≤10 condition at the same time. The rule is built to be rare.	\$5,000	+\$0	+0.00%
Vampire 5% One gated session exists. Range spans the pre-gate expectancy and today's gated result; neither is a forecast.	\$5,000	-\$1,017 to +\$286	range
Bars only	\$100,000	+\$1,085	+1.08%

With the CSP extrapolation the ending equity is \$103,960 (+3.96%); adding the Vampire's range gives \$102,944 to \$104,246. Max drawdown on the bars-only curve: -1.04%.

SIXFOLD, per name

HRB		+\$568	AAPL		+\$269
NVDA		+\$249	META		+\$214
ABBV		+\$213	MSFT		+\$207
V		+\$171	KO		+\$100
LLY		+\$88	PG		+\$70

COST		-\$52	FHI		-\$99
HD		-\$215	AMZN		-\$219
UTHR		-\$225	GOOGL		-\$254

\$3,750 per name, whole shares, bought at the August 4 open. \$17,969 of the sleeve sat idle in whole-share rounding and the four unused slots, which mirrors the live book.

WHAT WE DON'T CLAIM

The numbers a skeptic should push on, listed by us first.

Two sessions of regime-gate data is a sign, not proof.

$p = 0.20$ would need roughly three times the 49 windows we have before it could settle either way.

This is a paper account.

Nothing here has met a real fill, a real borrow or a real margin call. Seven days cannot tell skill from luck on any sleeve.

The 30-day simulation's CSP line is an extrapolation.

This plan has no historical option quotes, so the sleeve's realized premium from five live sessions is scaled to 23. Assignment risk is not modelled. The bars-only line is the one to trust.

The Vampire signals off Alpaca's free IEX feed and fills against the consolidated book.

Live fills landed 12 to 47 cents from that feed's mid on QQQ. A consolidated feed would change its numbers more than any gate.

CSP sits at about twice its 15% target.

Our own morning check hid that for two days behind a hardcoded threshold; it now reads the live config.

The advisor call fails.

When it does, the engine logs it, closes entries, and moves on. Every failure is in the chained journal.

Every requirement, and where the evidence lives.

Autonomous agent on the Trading API	alpaca-py for equities, options, IOC and DAY limit orders and closes; runs unattended on EC2 under systemd with a watchdog and a post-open verification timer.
MCP server or CLI	The Alpaca MCP server (<code>alpaca-mcp-server</code> over stdio) supplies live option quotes to the CSP scanner, which refuses to price any contract it cannot quote.
Options trading	Cash-secured puts on eight names (level 3), one covered call (KO \$93, Oct 2), a disposal guard so no automated path can naked the call.
Fresh \$100k paper account	PA3TPECBLC29, started at \$100,000 on August 28.
One-page write-up	docs/WRITEUP.md : AI logic, risk gates, Alpaca infrastructure, what we measured and cut, what we don't claim.
Public repository	github.com/mihai-satmarean/alpaca-trading-agents , 665 tests, 73 merged pull requests in seven days.
Application URL	Live dashboard at <code>http://44.202.110.127:8501/</code> behind a fail-closed access token supplied on the submission form: equity hero and session curve, sleeve budgets from the broker's book, positions ledger, AI Council, SIXFOLD scanner, options scanner, hash-verified notification journal.
Video and slides	This page is the deck. Sections map to the judging criteria: P&L, technology, creativity, presentation.

Team

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Figures on this page: live account as of September 3, 2026, 15:25 ET; simulation on Alpaca daily bars, August 4 to September 3, 2026, adjusted; Vampire gate scored on 2,828 real fills. Built by ProductAdvisors for the lablab.ai × Alpaca AI Trading Agents Hackathon.